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|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16 | D | Q leads P by $\frac{\lambda}{2}$ rad. Equivalently, P leads Q by $\frac{3\lambda}{2}$ rad.                                                                                                                                                                         |
| 17 | A | $\theta_{\min} \approx \frac{\lambda}{b} = \frac{550 \times 10^{-9}}{35 \times 10^{-2}} = 1.57 \times 10^{-6} \text{ rad}$ $\theta_s = \frac{s}{r} \Rightarrow s = r \theta_s = r \theta_{\min} = 200,000 \times 1.57 \times 10^{-6} = \underline{0.31 \text{ m}}$ |
| 18 | C | Using $d \sin \theta = n\lambda$<br>$(1/3.0 \times 10^5) \sin 90^\circ = n_{\max} (600 \times 10^{-9})$                                                                                                                                                            |